The following section will cover deploying a serverless application like Quote Function on a Kubernetes-based platform using Knative.

16.3 Deploying serverless applications with Knative

In the previous sections, you learned about Spring Native and how to use it with Spring Cloud Function to build serverless applications. This section will guide you through deploying Quote Function to a serverless platform on top of Kubernetes, using the Knative project.

Knative is a “Kubernetes-based platform to deploy and manage modern serverless workloads” (https://knative.dev). It’s a CNCF project that you can use to deploy standard containerized workloads and event-driven applications. The project offers a superior user experience to developers and higher abstractions that make it simpler to deploy applications on Kubernetes.

You can decide to run your own Knative platform on top of a Kubernetes cluster or choose a managed service offered by a cloud provider, such as VMware Tanzu Application Platform, Google Cloud Run, or Red Hat OpenShift Serverless. Since they are all based on open source software and standards, you could migrate from Google Cloud Run to VMware Tanzu Application Platform without changing your application code and with minimal changes to your deployment pipeline.

The Knative project consists of two main components: Serving and Eventing.